

P3 — Effect Modification / Interaction (Lab 4)

Model submission — demonstration paper (Kuch et al., BMJ Open 2026).

The candidate modifier and why we chose it

We test **sex** as an effect modifier of the phenotype–mortality association. Sex is the strongest candidate here for two reasons. First, it is clinically plausible: men and women differ in fat distribution, in the visceral-adiposity thresholds at which cardiometabolic risk rises, and in baseline all-cause mortality — so a WHtR-defined “high central adiposity” contrast could act differently across sexes. Second, and more decisively for us, **sex is entangled with the exposure itself**. The four phenotype groups are far from sex-balanced: Group II (obese, high central) is 56% female while Group I (obese, low central, our reference) is only 15% female. When the composition of the comparison groups shifts with sex, any crude between-group contrast partly reflects who is in each group rather than the effect of central adiposity — exactly the situation where a stratified analysis can mislead and a formal interaction test is required.

Diagnostic only — the retired IV-vs-I stratification

This table is not the P3 result. It uses **Group I as reference**, which is the paper’s contrast, not our locked estimand. It is reproduced here solely to show *why* an earlier draft reached a dramatic conclusion, and it is superseded by the targeted IV-vs-III analysis below. Read that one for the finding.

Refitting the crude Cox model within each sex against Group I (Table P3.1), the all-cause HRs are dramatically larger in men.

Table P3.1. Crude sex-stratified all-cause HR (95% CI), Group I = reference. *Locked domain, N = 6,048 — the rows P3_code fits. Group I holds 4 deaths: 2 of 317 men and 2 of 58 women.*

Group	Men	Women
II (obese, high central)	16.5	2.16
III (non-obese, low central)	12.2	2.28
IV (non-obese, high central)	36.1 (8.9–147.5)	4.11 (0.99–17.1)

Read naively, this is a striking picture: for the headline Group IV, the crude HR is nearly nine-fold higher in men (36.1) than in women (4.11), and the women’s interval even crosses 1. Taken at face value it would suggest that non-obese central adiposity is far more lethal for men.

The formal interaction test — targeted at the locked contrast

Our locked estimand is **IV vs III**. An earlier draft tested **group * sex** across all four groups, which is a **multi-degree-of-freedom test of whether any group contrast varies by sex** — a broader question than ours. Its p-values (0.055 crude, 0.571 adjusted) answered that broader question and are **retired**, not carried forward.

To test our estimand we restrict to Groups III and IV, making the interaction term exactly **1 df for the contrast we care about**. Crude and adjusted are fit on the **same Model-1 complete-case rows** (n = 1,296 of 1,306; 937 men, 359 women), so any difference between them is attributable to adjustment rather than to sample change.

	ratio of HRs (women / men)	p
Crude	0.62 (0.26–1.43)	0.278
Model 1 adjusted	1.00 (0.42–2.35)	0.993

Sex-specific adjusted IV vs III hazard ratios (Model 1, fitted within each sex):

	n	HR (95% CI)
Men	937	1.10 (0.75–1.62)
Women	359	1.08 (0.49–2.38)

The additive scale — a real estimand, not an inference from hazard ratios

Stratum-specific hazard ratios do not show you the additive picture; you have to estimate it. We take a **six-year horizon** (natural given the paper’s ~6.9-year mean follow-up) and compute **standardized risks by g-computation**: fit the Model-1 Cox model within each sex, then predict six-year risk for *every* person in that stratum twice — once setting IV = 0, once setting IV = 1 — and average. That marginalizes over the observed covariate distribution of that stratum, so the contrast is a population-average risk difference rather than a covariate-conditional one. Confidence intervals are from 300 bootstrap resamples within stratum.

	6-yr risk, Group III	6-yr risk, Group IV	Risk difference
Men	0.097	0.105	+0.008 (–0.033 to 0.042)
Women	0.101	0.106	+0.005 (–0.073 to 0.076)

Difference in risk differences (women – men): **–0.003**.

Interpretation

What we can say. On the multiplicative scale, the adjusted ratio of hazard ratios is 1.00, and on the additive scale both six-year risk differences are under one percentage point. The data are **compatible with little or no interaction by sex** for our locked contrast.

What we cannot say. The confidence interval on that ratio runs from **0.42 to 2.35** — compatible with the effect in women being less than half that in men, or more than twice it. So the data are *also* compatible with substantial heterogeneity. “No evidence of multiplicative interaction **at the precision available**” is the honest statement; “no effect modification” is not. With 359 women and few events in the non-obese strata, this study is not powered to resolve the question.

What we specifically withdraw. An earlier draft said the crude difference was “confounding by baseline risk, not true modification” and that the adjusted test “refuted” modification. Neither follows. A p-value that moves when covariates enter tells you the *estimate* changed; it does not establish **what generated** the crude difference, and a non-significant test is not evidence of absence. The crude ratio of 0.62 and the adjusted ratio of 1.00 are both imprecise; the change between them is within their joint uncertainty.

Method note — what we did and did not avoid

We did **not** run a logistic RERI. The problem is specific and worth stating correctly: the error is analysing *ever-died* with logistic regression, which discards follow-up time and ignores censoring, so any additive measure built on it answers a question about cumulative status at an unspecified time. **RERI is not inherently a logistic-regression concept** — additive interaction can be estimated on a survival scale. Our six-year standardized risk differences above *are* the additive-scale analysis, done in a way that respects time-to-event.

Two limitations of the additive estimate. Several bootstrap replicates in the female stratum triggered Cox convergence warnings — with 359 women and sparse events, some resamples separate — so the female interval is wide and should be read as order-of-magnitude. And all of this is **unweighted**: P3 precedes the survey-design increment. P4 shows that design materially changes inference for the main estimate, so whether the design-aware interaction gives the same qualitative answer is an open question, and one M1 should check rather than assume.

Bottom line

For the locked IV-vs-III contrast, sex does not show evidence of effect modification on either scale at the precision this study affords — adjusted ratio of hazard ratios 1.00 (0.42–2.35), six-year risk differences +0.008 in men and +0.005 in women. The eye-catching crude stratified gap reported by an earlier draft came from a different contrast (IV vs I) and a broader test. This is a null reported with its uncertainty attached, not a demonstration that sex is irrelevant.